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Saveetha Institute of Medical And Technical Sciences
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Design And Development Of A Direct Farmer To Customer Marketplace System

CSA0915-Programming In Java

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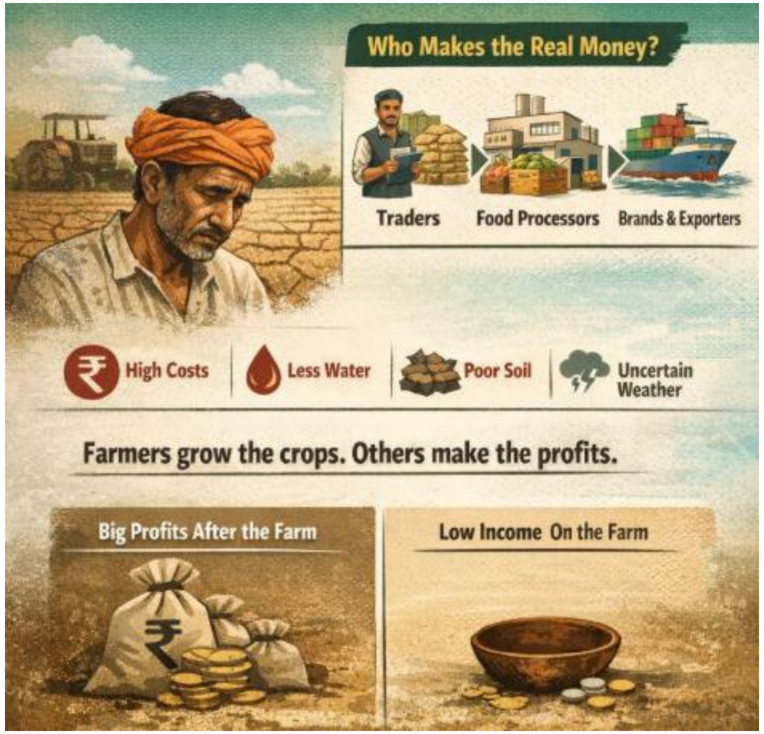
INTRODUCTION

- Agriculture is a key sector, but farmers face difficulties in getting fair prices due to middlemen.
- Consumers often pay higher prices while farmers receive less profit.
- A Direct Farmer-to-Consumer Marketplace System helps connect farmers and buyers directly.
- This platform allows farmers to sell products online and reach more customers easily.
- The system improves transparency, increases farmers' income, and provides fresh products to consumers.

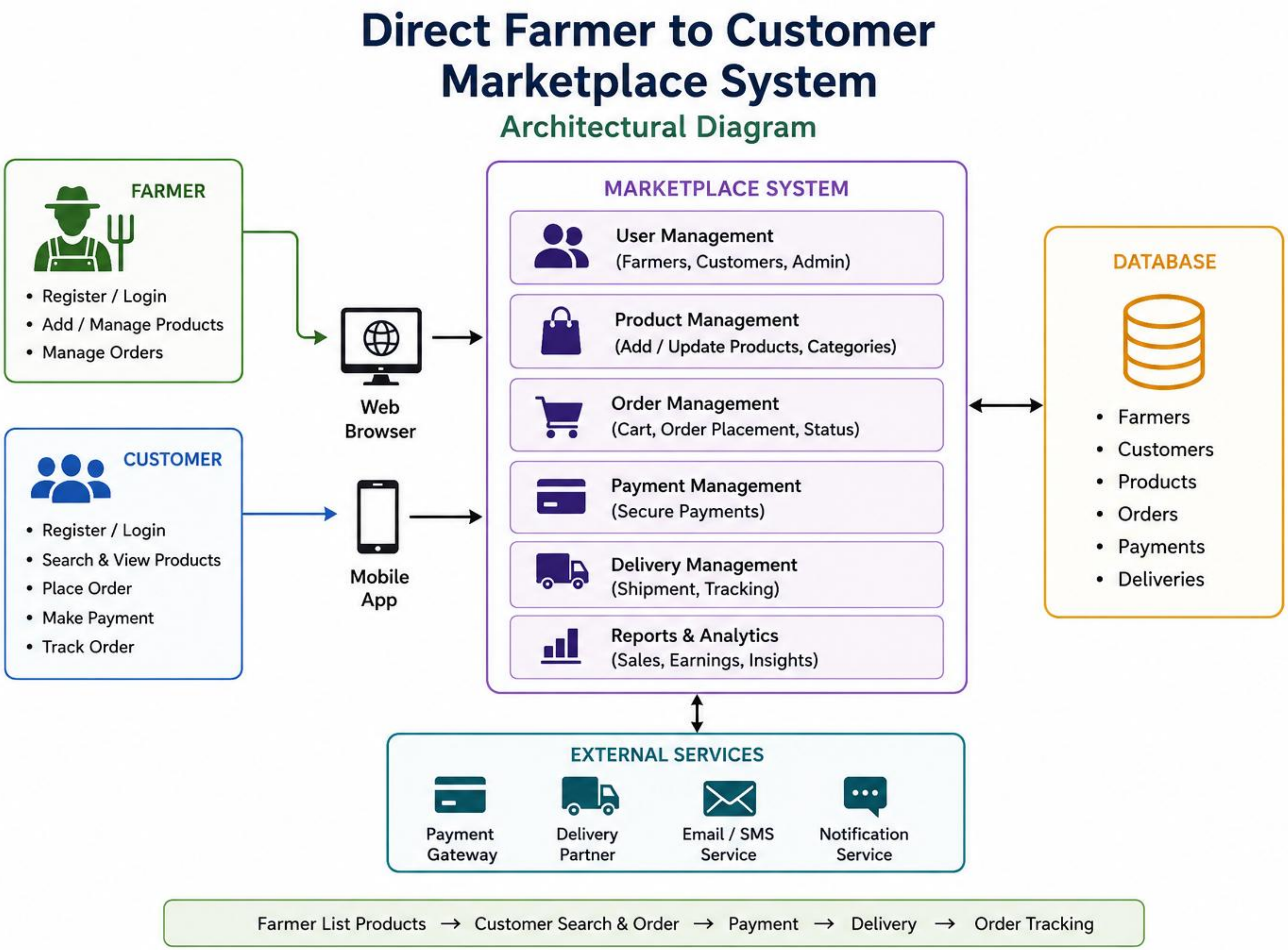
PROBLEM STATEMENT

- Farmers receive low profits due to the involvement of multiple intermediaries in the supply chain.
- Consumers pay higher prices for agricultural products because of added middlemen costs.
- Lack of direct communication between farmers and consumers leads to inefficiencies and mistrust.
- Farmers do not have access to a reliable platform to sell their products to a wider market.
- Traditional systems cause delays, wastage of perishable goods, and lack transparency in pricing.



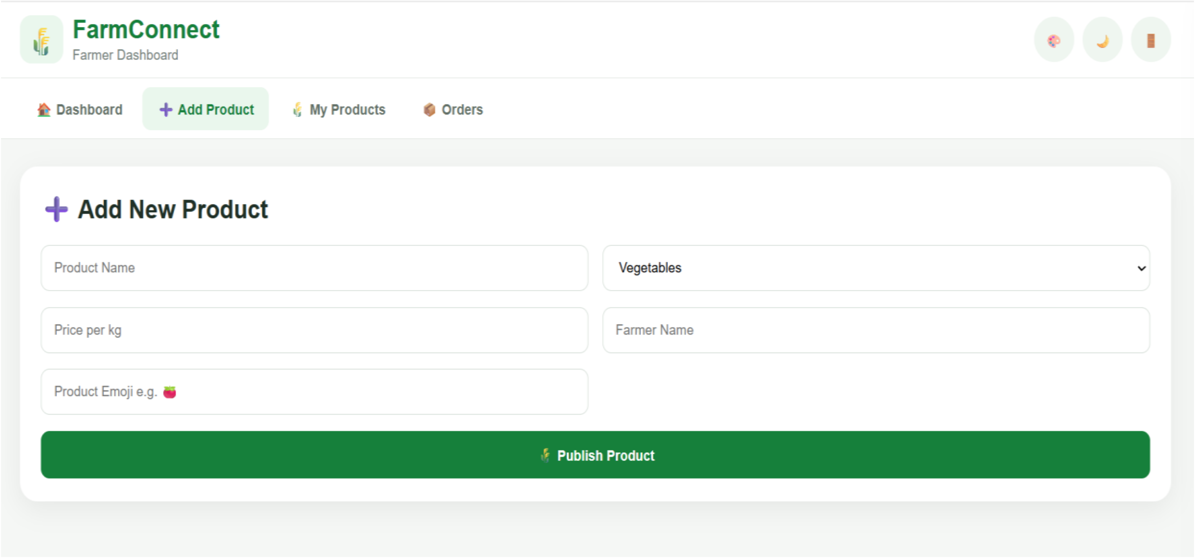
PROPOSED SYSTEM

- Farmers can register, add products with price, quantity, and update them anytime.
- Consumers can search, view, and purchase fresh products directly from farmers.
- Direct communication is enabled to improve trust and ensure price transparency.
- The system provides secure payments and simple order management features.
- Delivery tracking and admin monitoring ensure smooth and efficient operations.


Direct Farmer to Customer
Marketplace System
Architectural Diagram
FARMER
• Register / Login
• Add / Manage Products
• Manage Orders
CUSTOMER
• Register / Login
• Search & View Products
• Place Order
• Make Payment
• Track Order
Web Browser
Mobile App
MARKETPLACE SYSTEM
User Management
(Farmers, Customers, Admin)
Product Management
(Add / Update Products, Categories)
Order Management
(Cart, Order Placement, Status)
Payment Management
(Secure Payments)
Delivery Management
(Shipment, Tracking)
Reports & Analytics
(Sales, Earnings, Insights)
DATABASE
• Farmers
• Customers
• Products
• Orders
• Payments
• Deliveries
EXTERNAL SERVICES
Payment Gateway
Delivery Partner
Email / SMS Service
Notification Service
Farmer List Products → Customer Search & Order → Payment → Delivery → Order Tracking

Module 1: Farmer Registration & Product Management

- This module is responsible for allowing farmers to register on the marketplace and manage the agricultural products they want to sell directly to customers.
- The marketplace allows farmers to register themselves and directly list their products, making their products visible to customers.



Purpose:

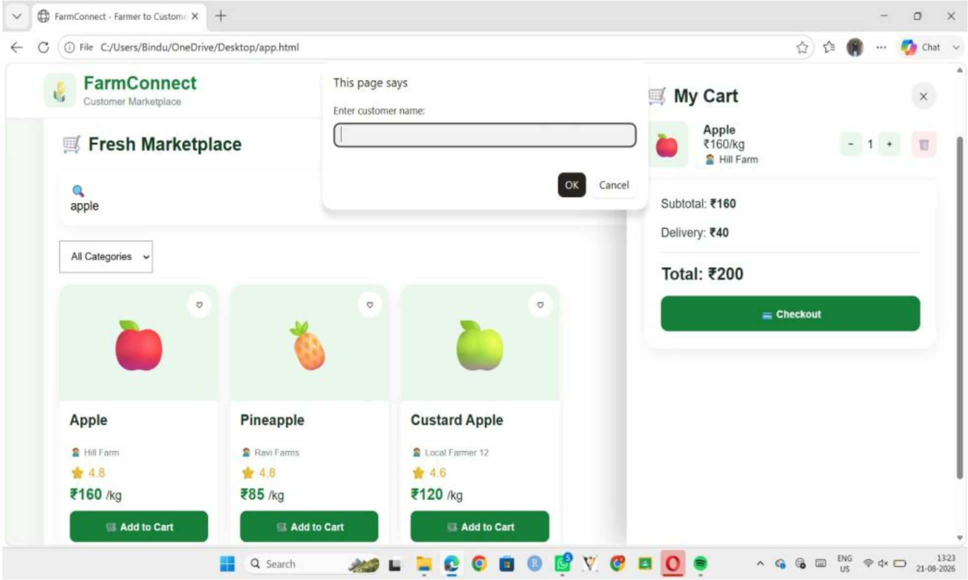
To provide farmers with a simple digital platform to sell their products directly to customers without depending completely on intermediaries.

Output

```
package addproducts;

public class AddProducts {

    public void addProduct(String name, double price, int quantity) {

        System.out.println("===== ADD PRODUCT =====");
        System.out.println("Product Name : " + name);
        System.out.println("Price        : Rs." + price);
        System.out.println("Quantity     : " + quantity);
        System.out.println("Product added successfully!");
    }
}
```

Module 2: Customer Product Search & Online Ordering

- Product Search – Customers can search for products using names, categories, brands, or keywords.
- Product Details – Customers can view the product name, price, description, images, and ratings.
- Add to Cart – Customers can select products and add them to the shopping cart.
- Online Ordering – Customers can enter delivery details, choose a payment method, and place the order.
- Order Confirmation & Tracking – The system provides an order ID and allows customers to track their order status.

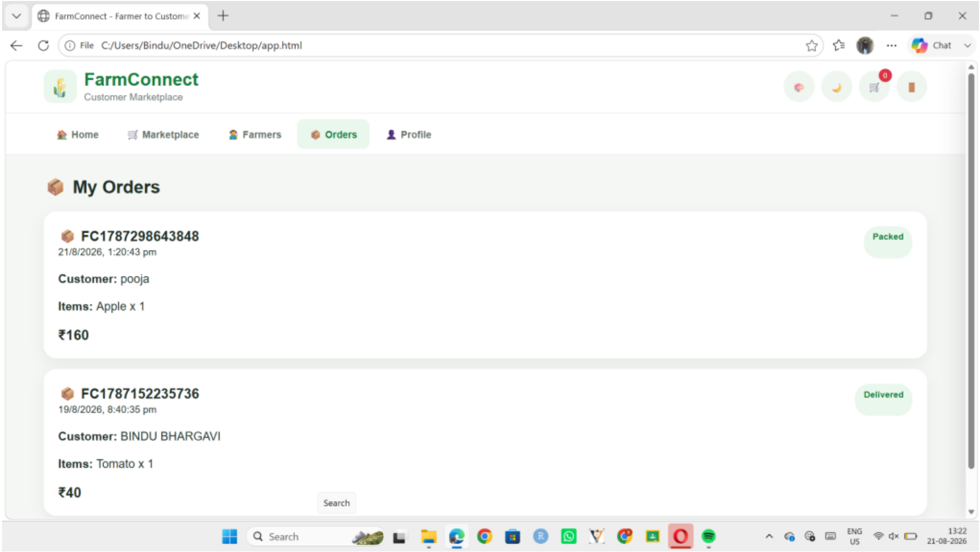


Output

```
package myproduct;

public class MyProducts {

    public void displayProducts() {

        System.out.println("===== MY PRODUCTS =====");

        System.out.println("1. Tomatoes");
        System.out.println("   Price    : Rs.50");
        System.out.println("   Quantity : 100");

        System.out.println();

        System.out.println("2. Potatoes");
        System.out.println("   Price    : Rs.40");
        System.out.println("   Quantity : 80");
    }
}
```

Module 3: Order, Payment & Delivery Management

- Order Management – The system manages customer orders, including order details, quantity, price, and order status.
- Payment Processing – Customers can select a suitable payment method and complete the payment securely.
- Order Confirmation – After successful payment, the system confirms the order and generates an order ID.
- Delivery Management – The system manages delivery details such as delivery address, delivery partner, and expected delivery date.
- Order Tracking – Customers can track their order status from Confirmed → Shipped → Out for Delivery → Delivered.



Output

```java
package orders;

public class Orders {

    public void displayOrders() {

        System.out.println("===== ORDERS =====");

        System.out.println("Order ID : 101");
        System.out.println("Product  : Tomatoes");
        System.out.println("Quantity : 5");
        System.out.println("Amount   : Rs.250");
        System.out.println("Status   : Delivered");

        System.out.println();

        System.out.println("Order ID : 102");
        System.out.println("Product  : Potatoes");
        System.out.println("Quantity : 10");
        System.out.println("Amount   : Rs.400");
        System.out.println("Status   : Processing");
    }
}
```

Future Scope

- Mobile Application – Develop a mobile app so farmers and customers can easily access the marketplace.
- Online Payment Integration – Add secure digital payment options for faster and easier transactions.
- Real-Time Delivery Tracking – Allow customers to track their orders and deliveries in real time.
- AI-Based Recommendations – Recommend suitable products to customers based on their search and purchase history.
- Expansion of Products – Include more farmers, agricultural products, organic products, and local produce.

Conclusion

The Direct Farmer-to-Customer Marketplace System provides a platform that connects farmers directly with customers. It helps farmers sell their products without depending heavily on intermediaries, while customers can purchase fresh products conveniently. The system supports product search, online ordering, payment, delivery, and order tracking. Overall, it can improve the efficiency of agricultural product sales and create a more direct and transparent connection between farmers and customers.

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